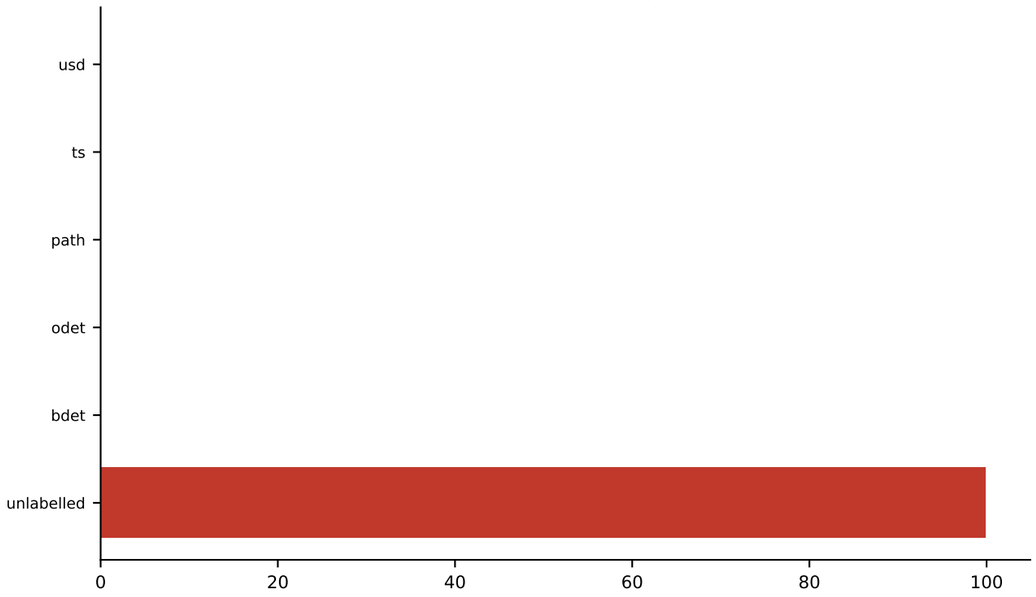
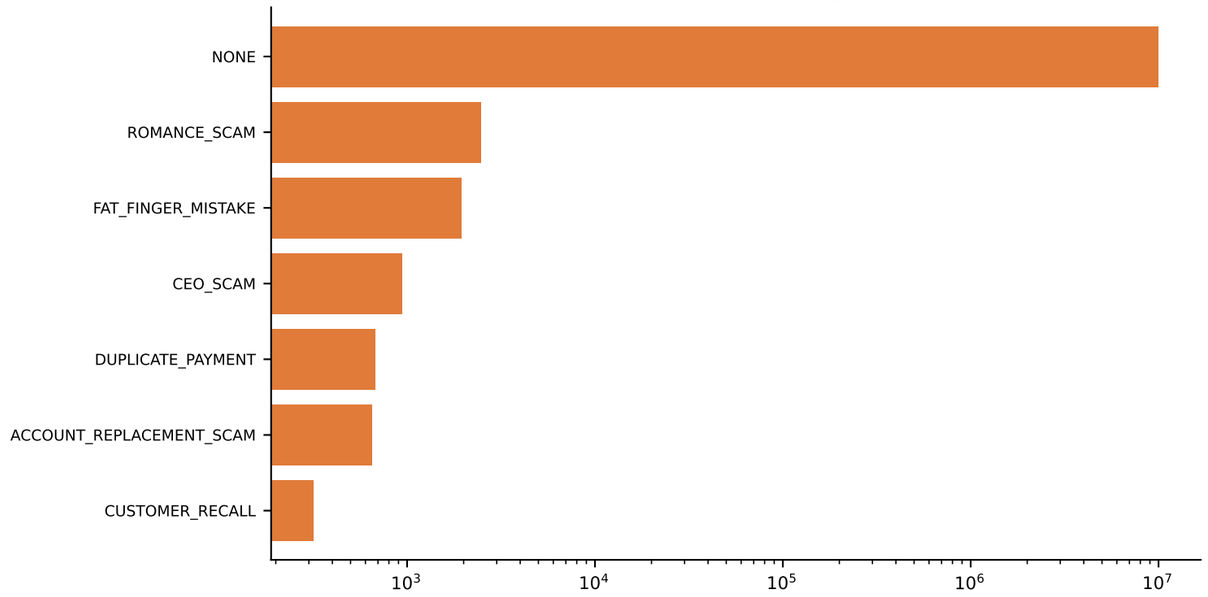
1. Data quality & scale

RULE / HYPOTHESIS: H: DISCLAIMER row + malformed rows are the only junk; no duplicate rows. Record counts, null rates and anomaly prevalence shown as baseline.

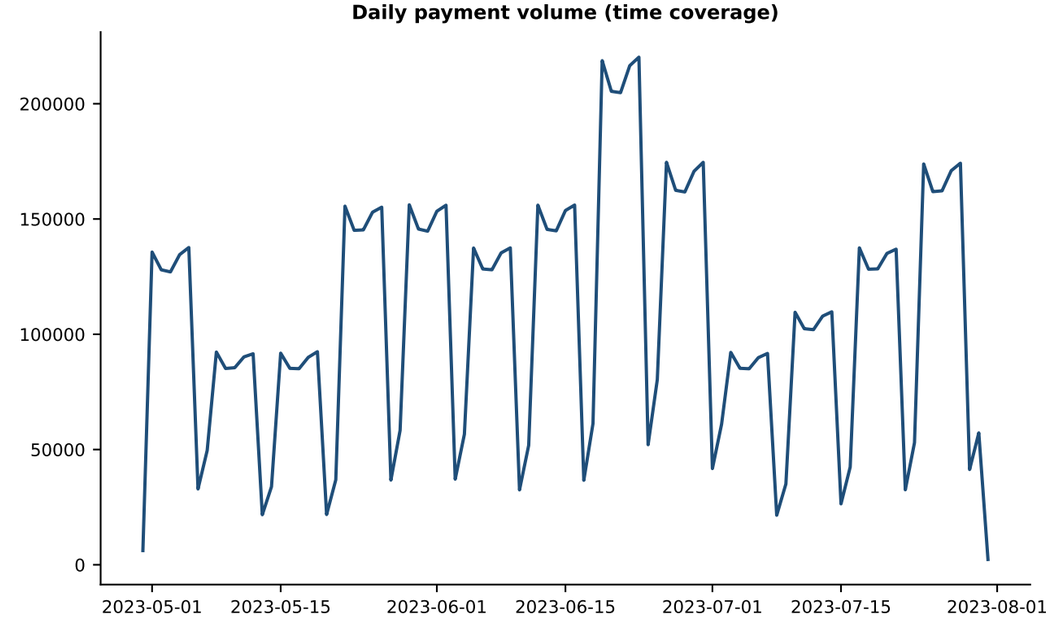
Share of rows with NULL (%) - anomaly NULL = normal traffic



Anomaly label counts (log scale)



Daily payment volume (time coverage)



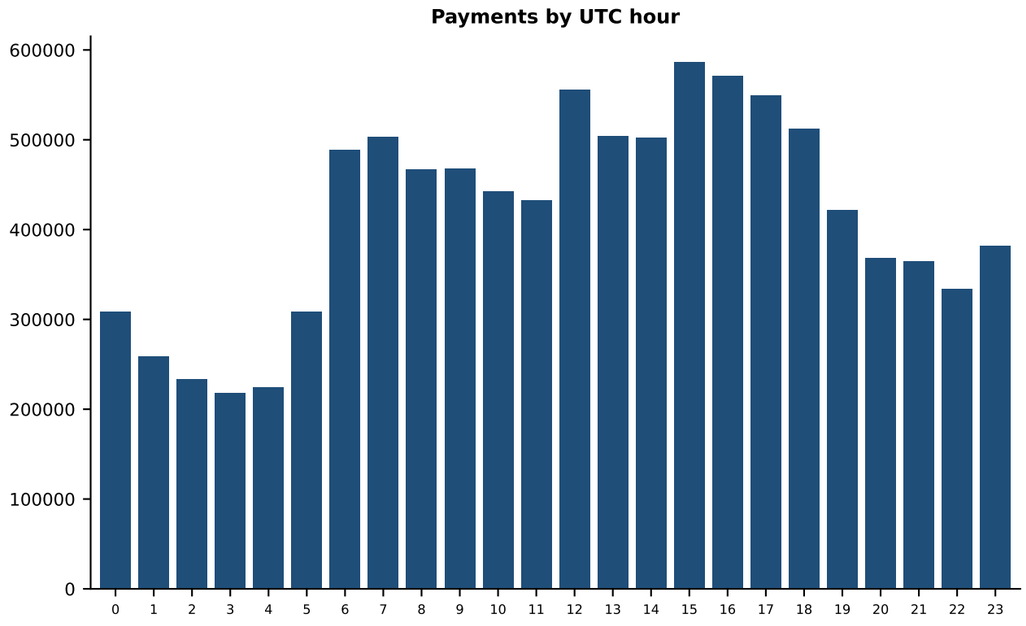
Integrity checks

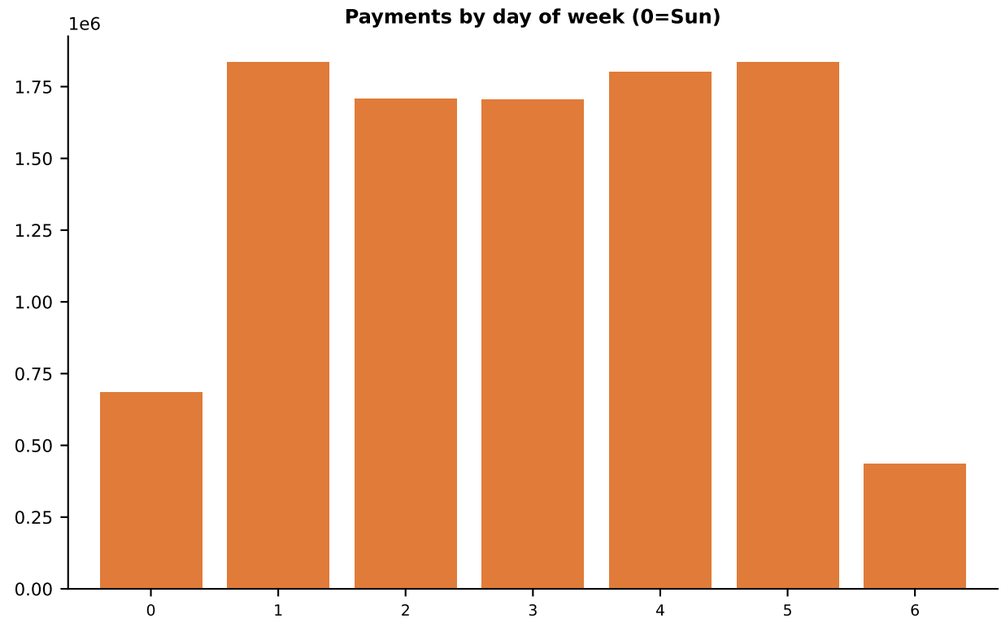
Rows: 10,000,000
Exact-duplicate rows: 1
Date range: 2023-04-30 to 2023-07-31
Zero/negative USD: 0

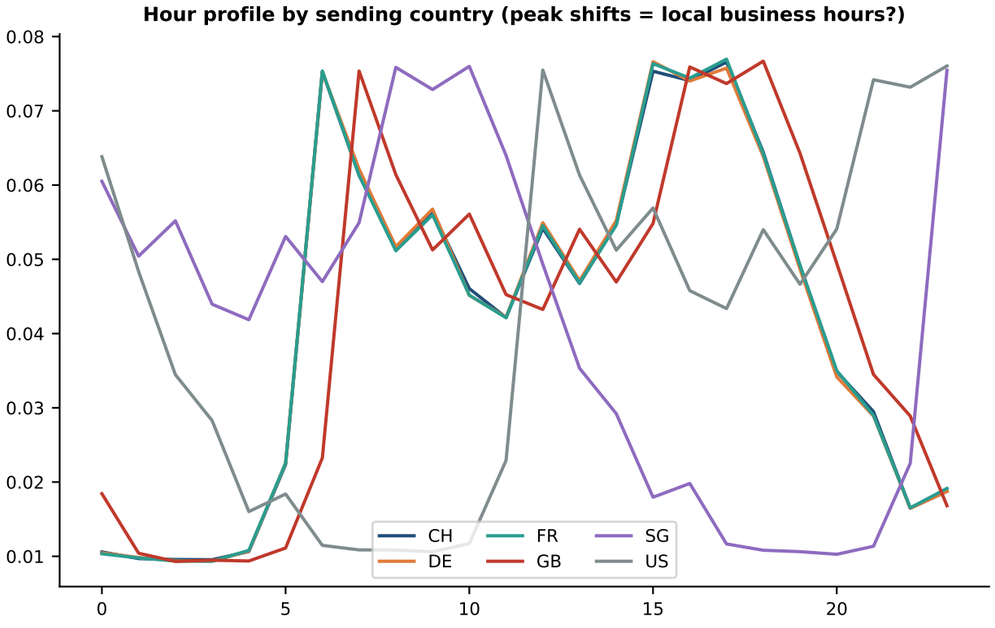
Source: Swift artificial payments dataset (n=10,000,000). All charts computed on the full data.

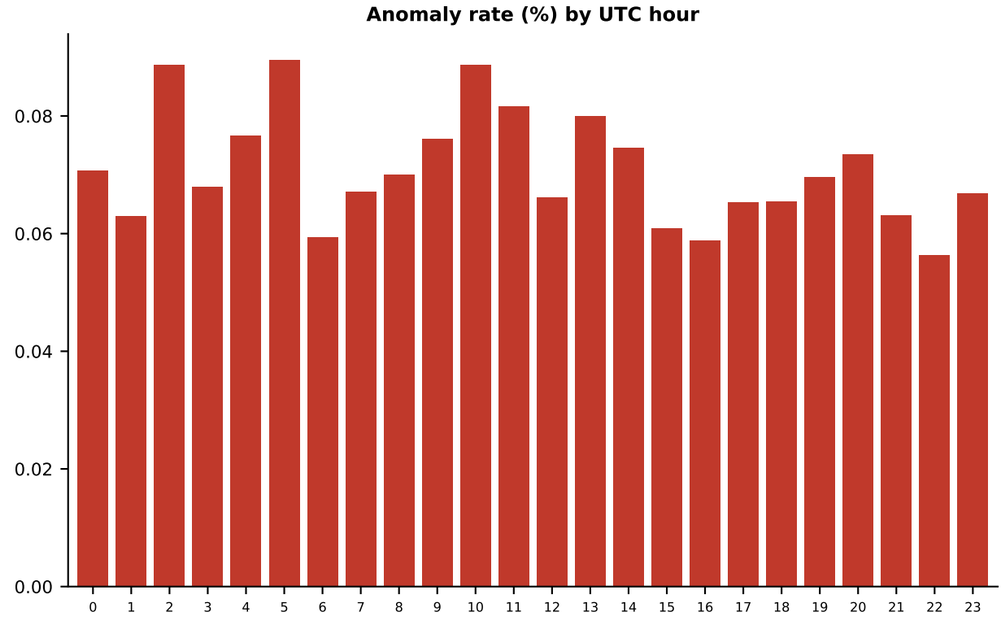
2. Time patterns

RULE / HYPOTHESIS: H: Volume follows business hours of the SENDER's country (UTC hour shifts by country); weekday >> weekend; anomalies may cluster off-hours.





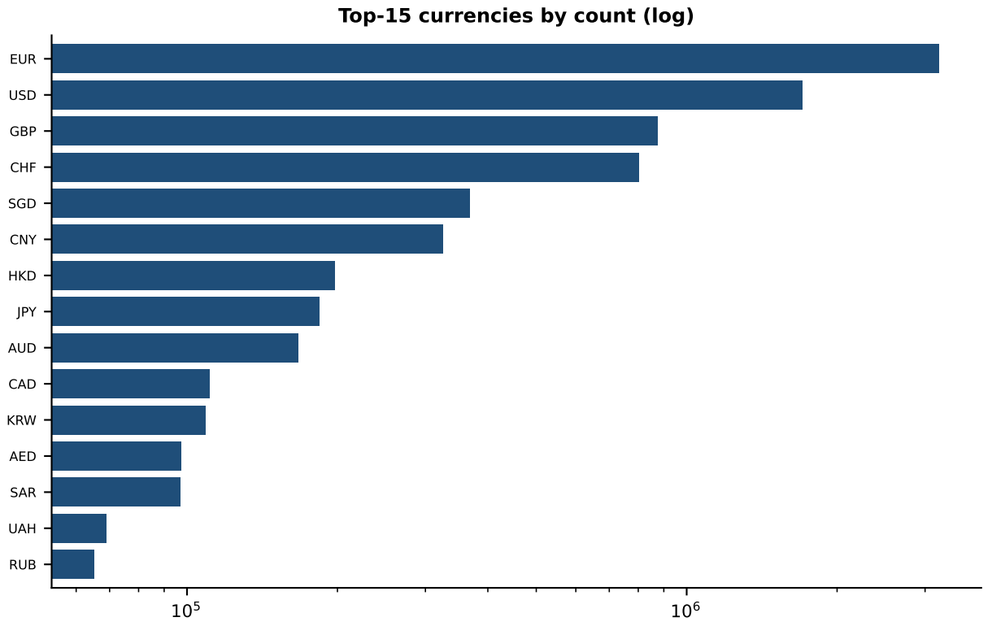


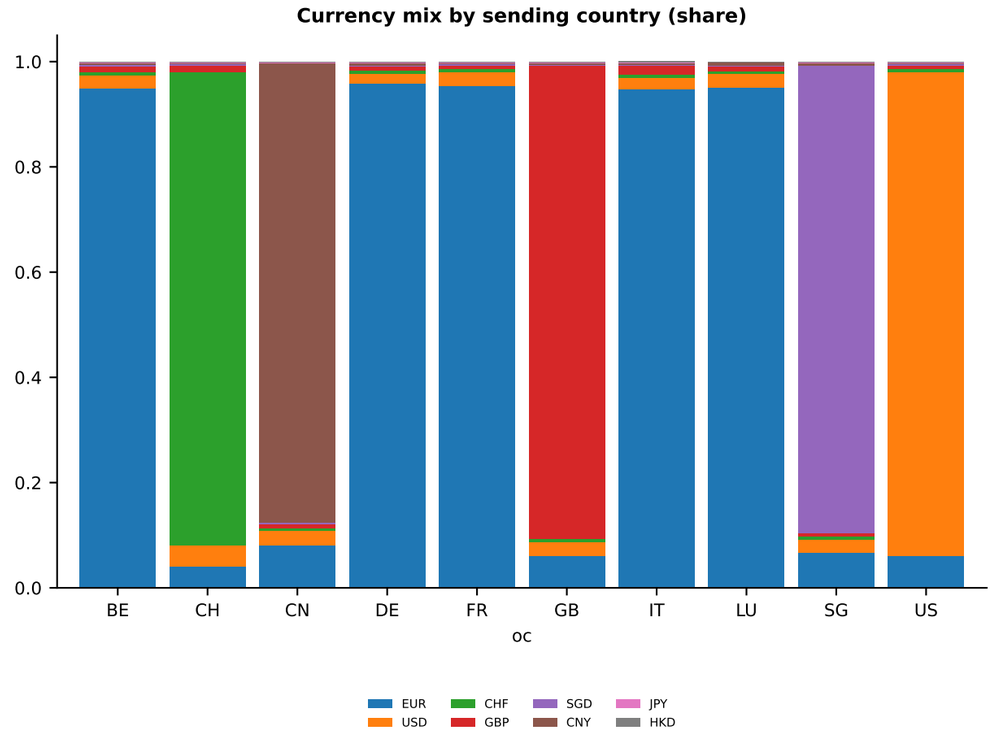


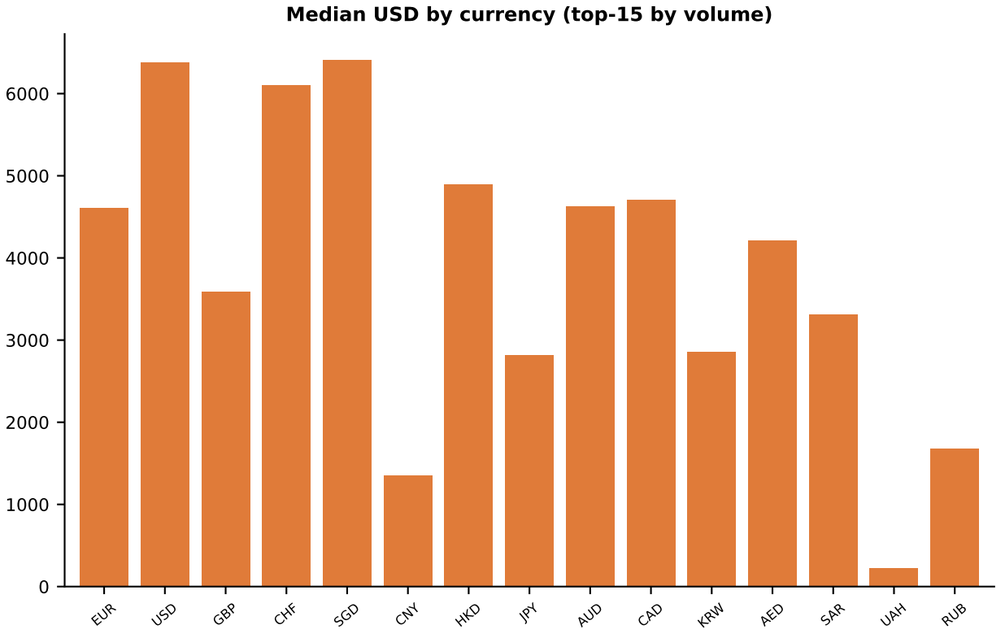
Source: Swift artificial payments dataset (n=10,000,000). All charts computed on the full data.

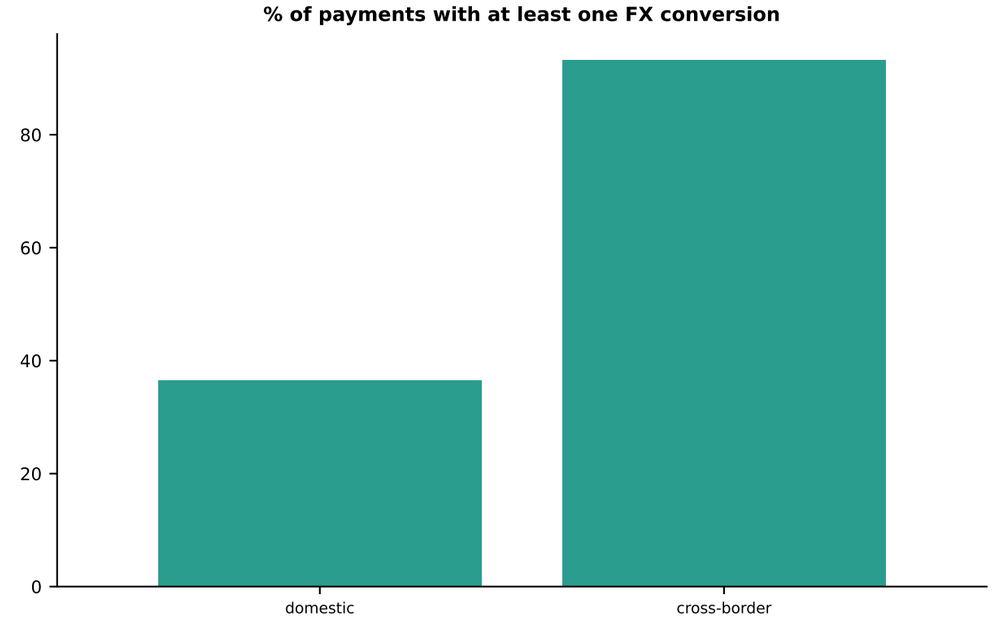
3. Currency

RULE / HYPOTHESIS: H: Currency is driven by the beneficiary/ordering country (EUR for euro-zone BICs); domestic payments rarely FX; a few currencies dominate volume.







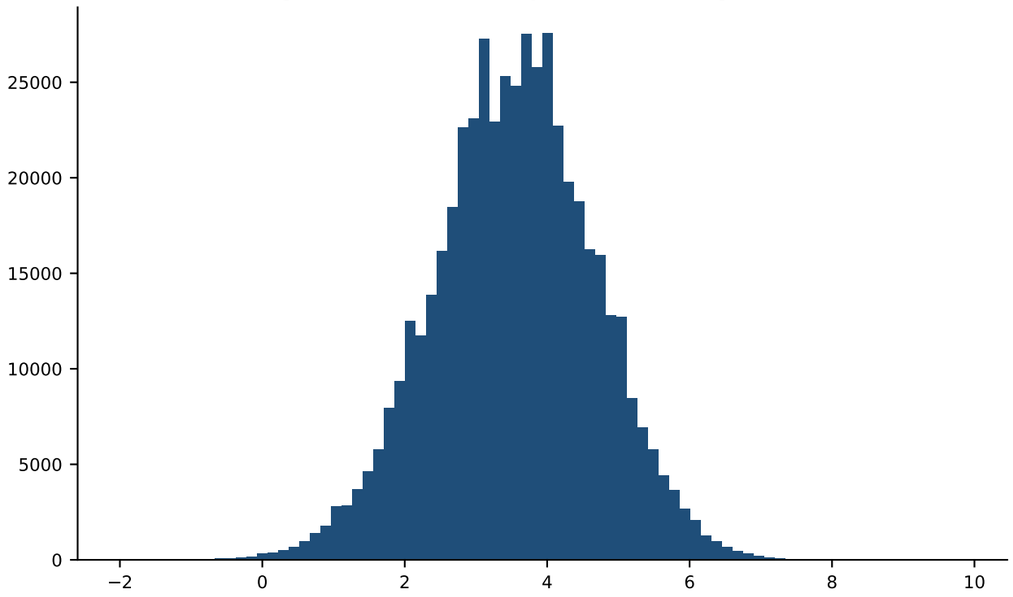


Source: Swift artificial payments dataset (n=10,000,000). All charts computed on the full data.

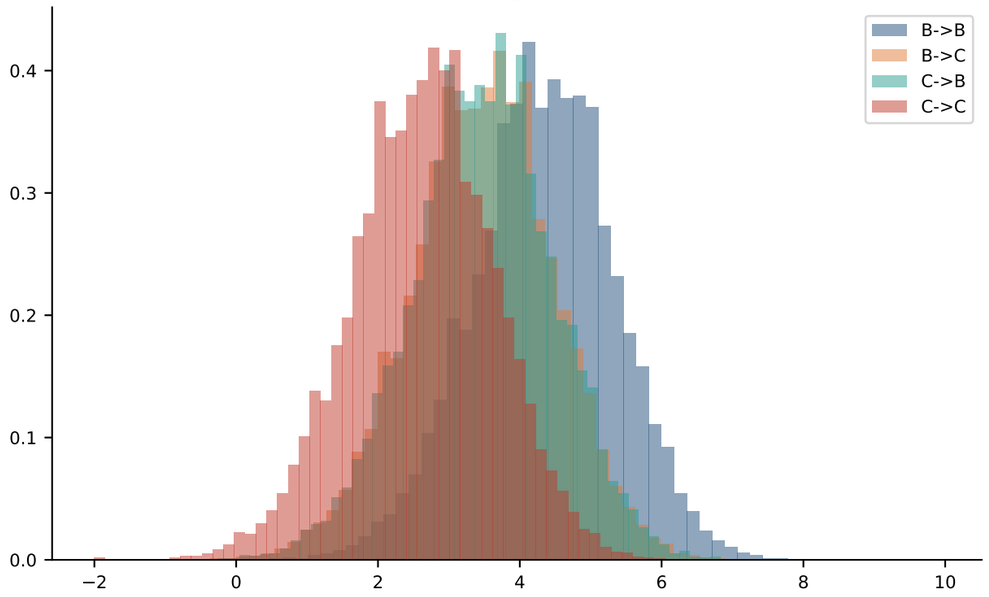
4. Payment value

RULE / HYPOTHESIS: H: Amounts are log-normal; scale depends on currency, B vs C party type and corridor; Business >> Consumer.

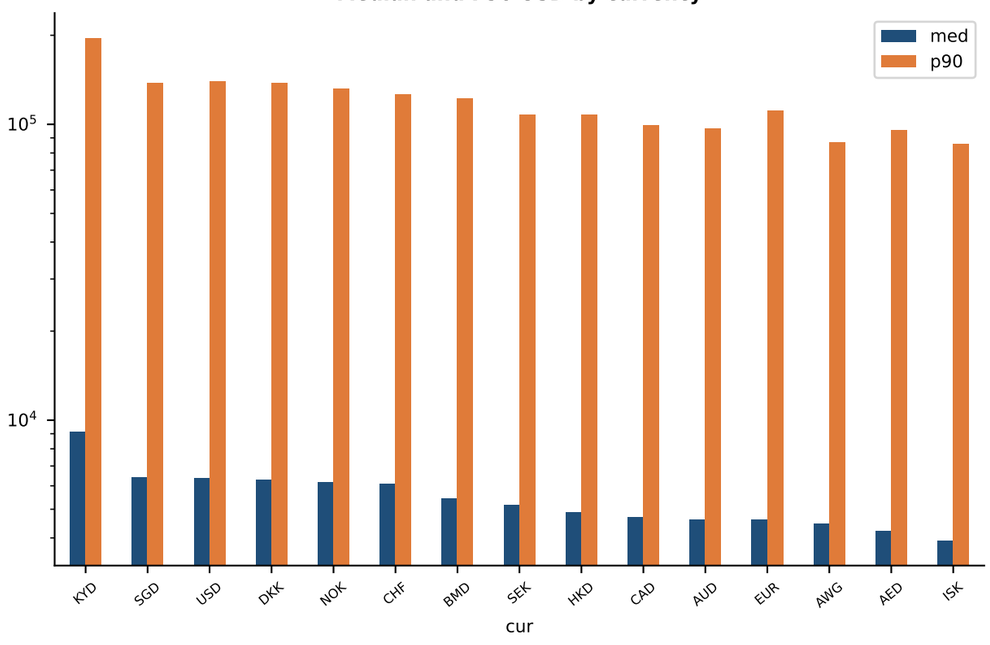
log10(USD amount): shape (normal => log-normal)



Amount by ordering->beneficiary type



Median and P90 USD by currency



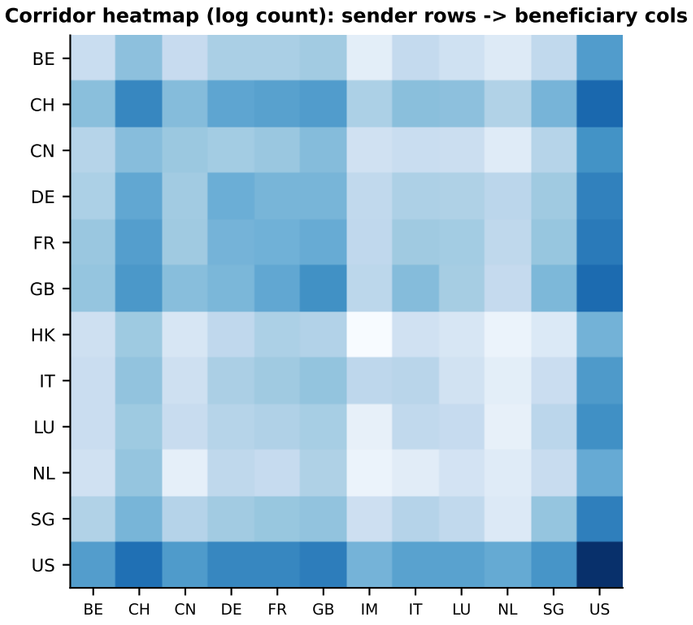
Round-number amounts (top values)

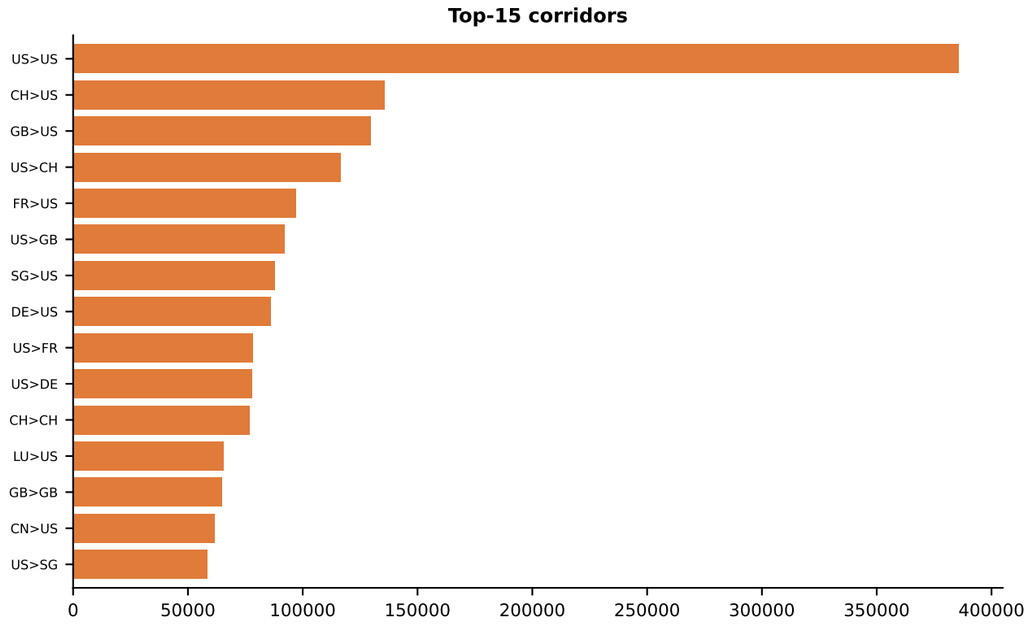
f	n
10000.0	35650
1000.0	32630
5000.0	21709
100000.0	16995
500.0	15544
100.0	13723
50000.0	13129
2000.0	5534
20000.0	4553
3000.0	4382
2500.0	4250
4000.0	3442
25000.0	3424
30000.0	3193
200.0	3016

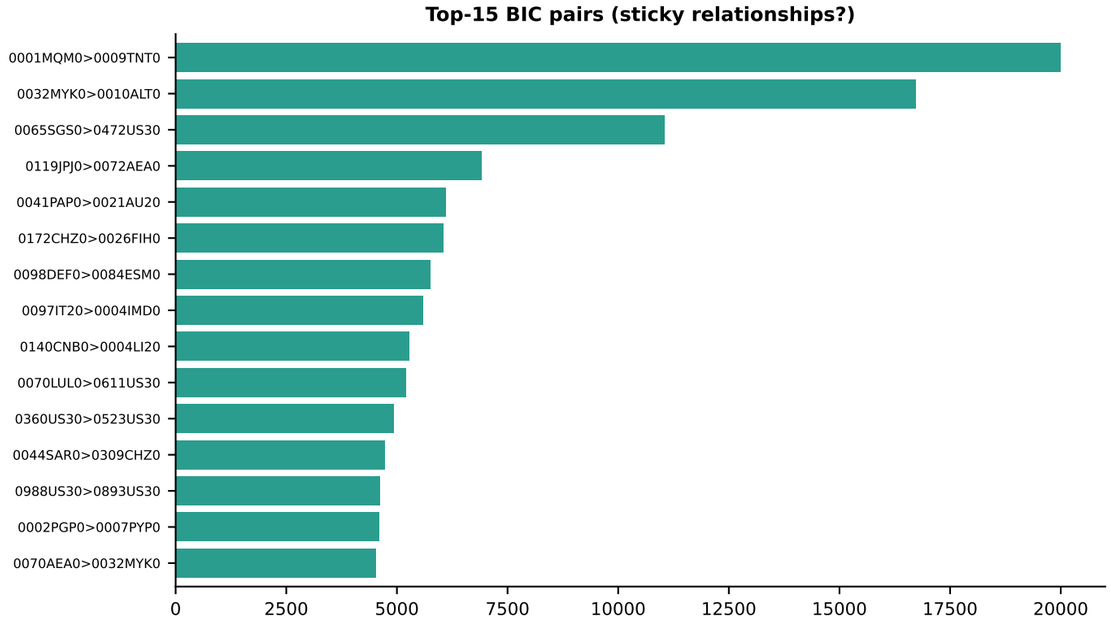
Source: Swift artificial payments dataset (n=10,000,000). All charts computed on the full data.

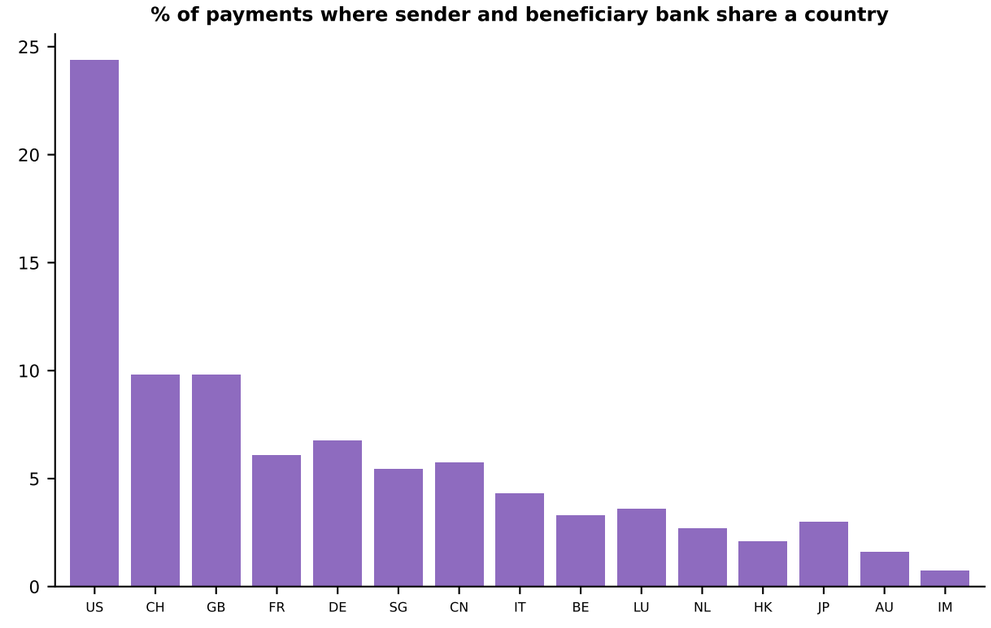
5. Routing & corridors

RULE / HYPOTHESIS: H: Corridors (ordering country -> beneficiary country) are concentrated; ordering/beneficiary BIC pairs are sticky.







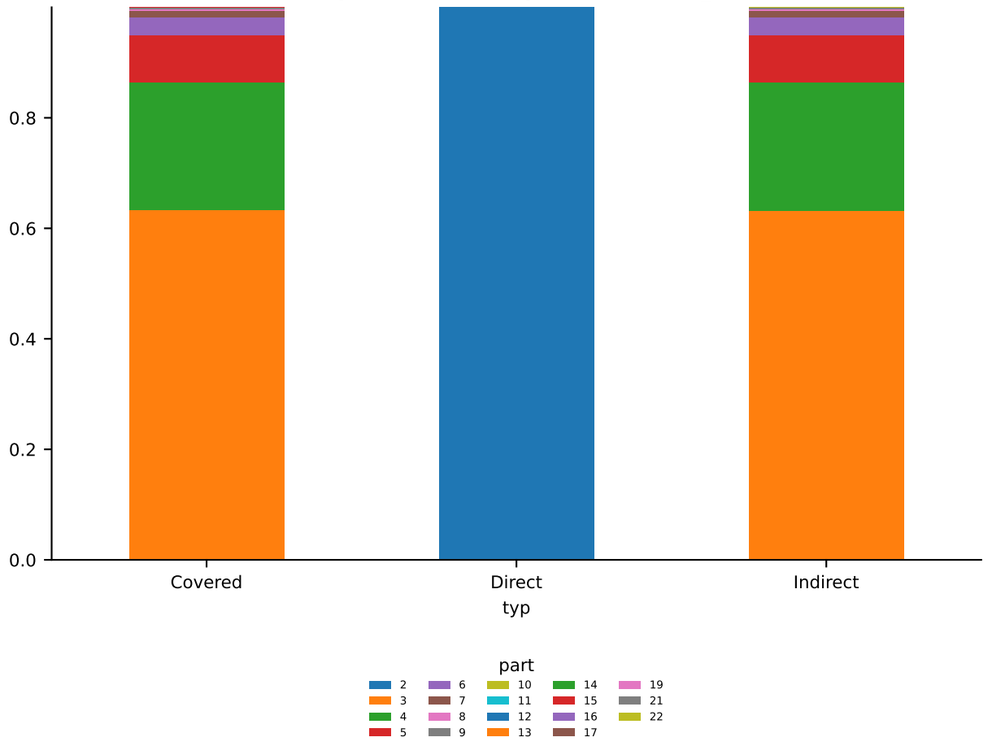


Source: Swift artificial payments dataset (n=10,000,000). All charts computed on the full data.

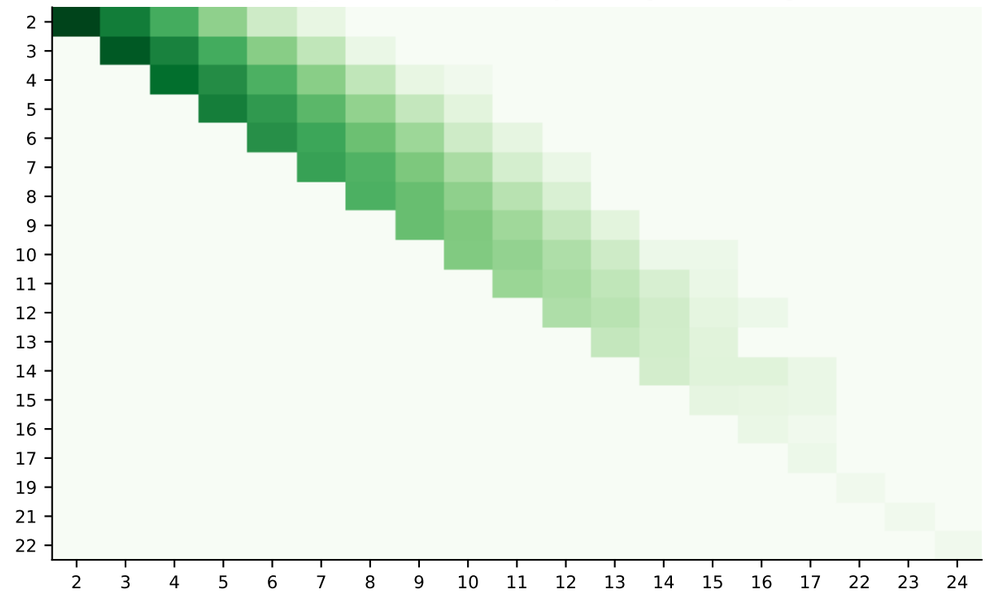
6. Path structure

RULE / HYPOTHESIS: H: TRANSACTION_TYPE determines PARTICIPANT_COUNT (Direct=2); path length = participants; FX count rises with participants & cross-currency corridors.

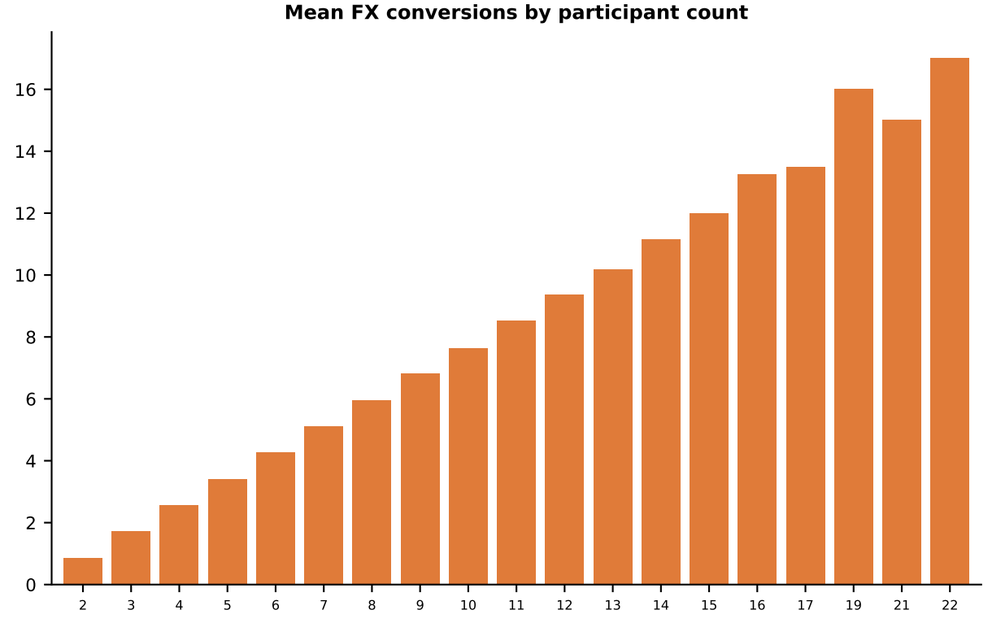
Participant count mix by TRANSACTION_TYPE



PARTICIPANT_COUNT (rows) vs BIC path length (cols), log count



Mean FX conversions by participant count



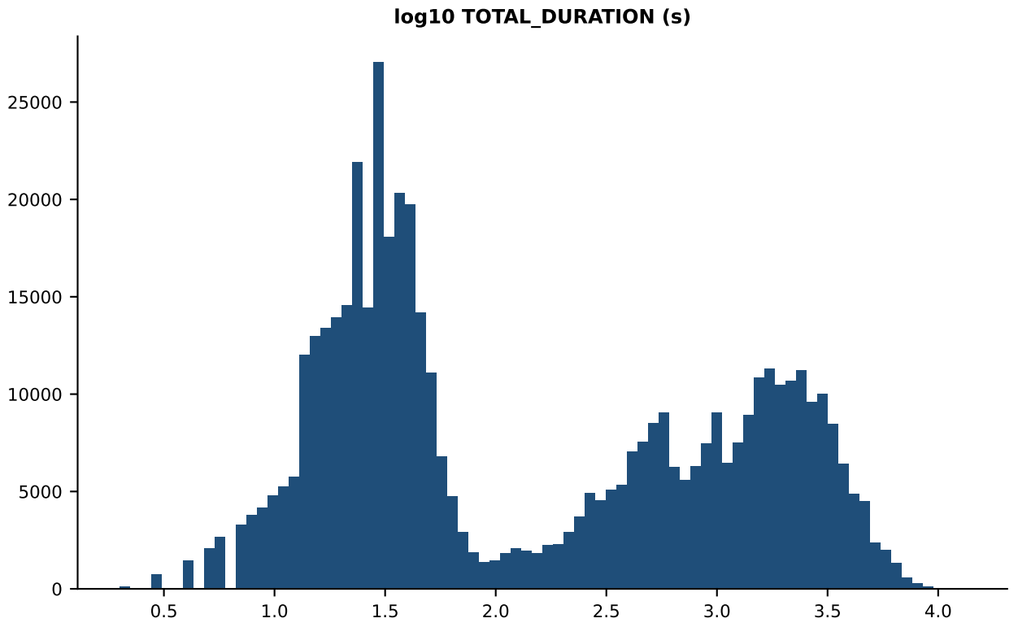
CONTINUITY_QFLAG vs '...' in path (should be perfectly aligned)

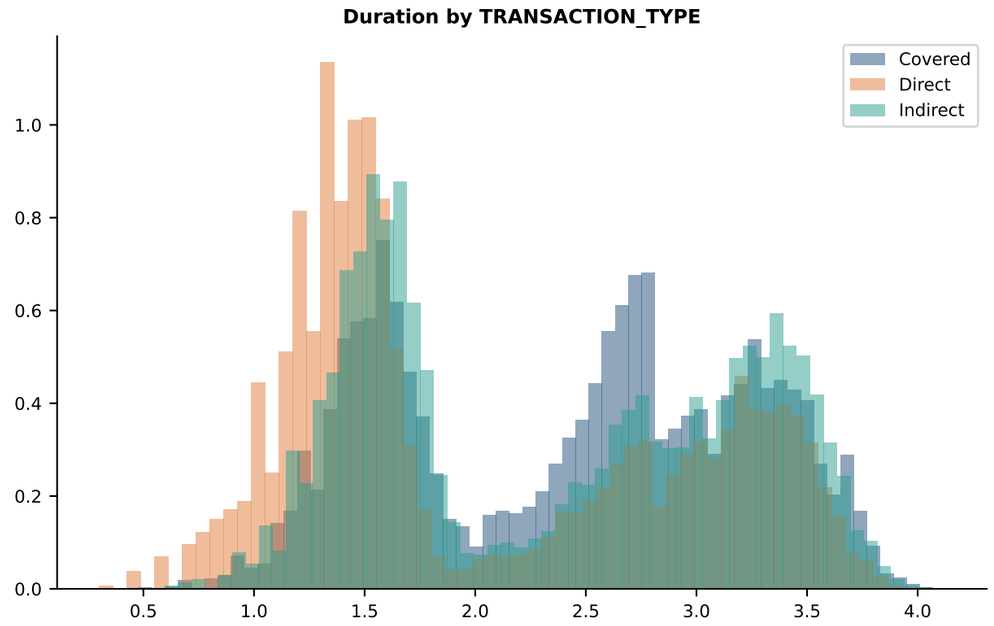
cont	path_gap	n
N	True	767204
Y	False	9232796

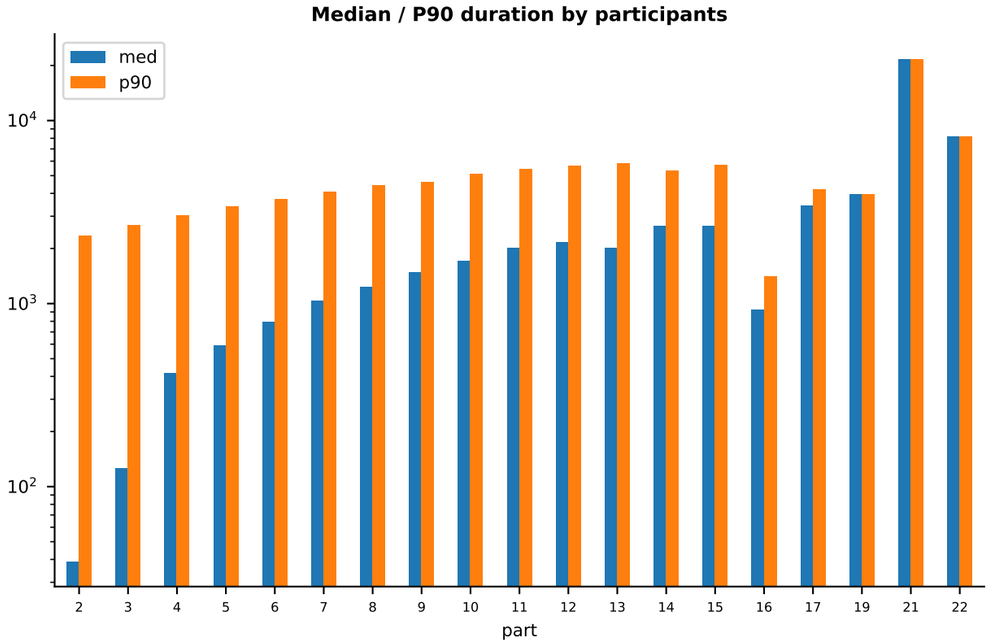
Source: Swift artificial payments dataset (n=10,000,000). All charts computed on the full data.

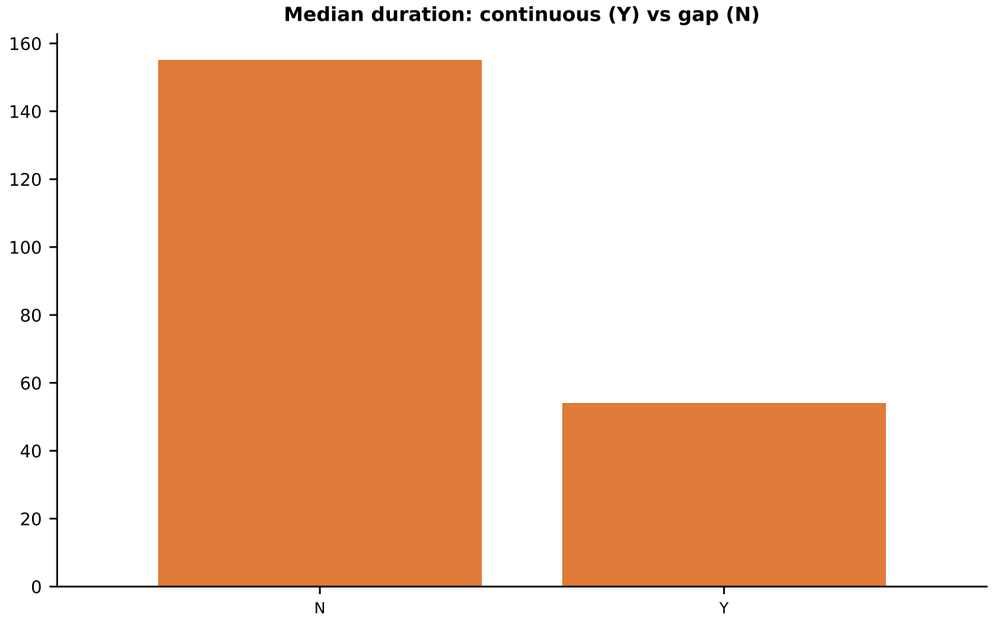
7. Duration

RULE / HYPOTHESIS: H: TOTAL_DURATION grows with participants and across time zones; Covered/Indirect slower than Direct; heavy right tail.





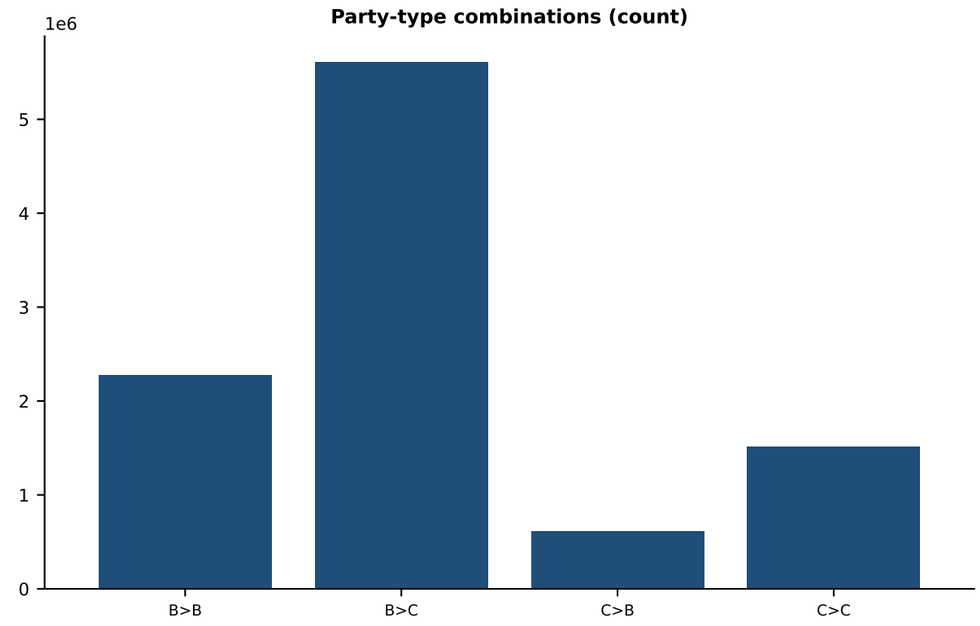


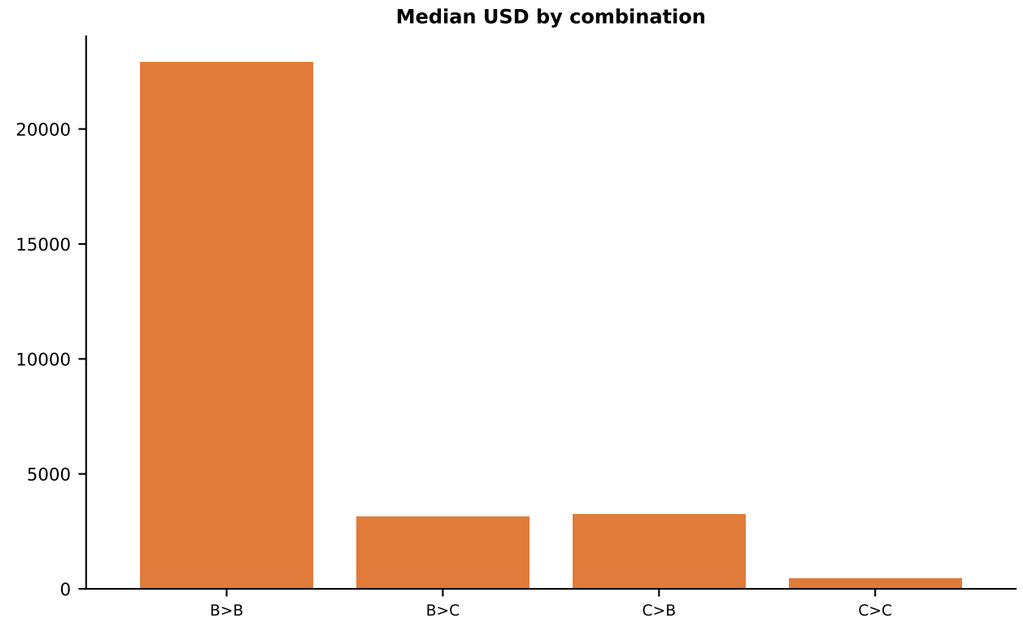


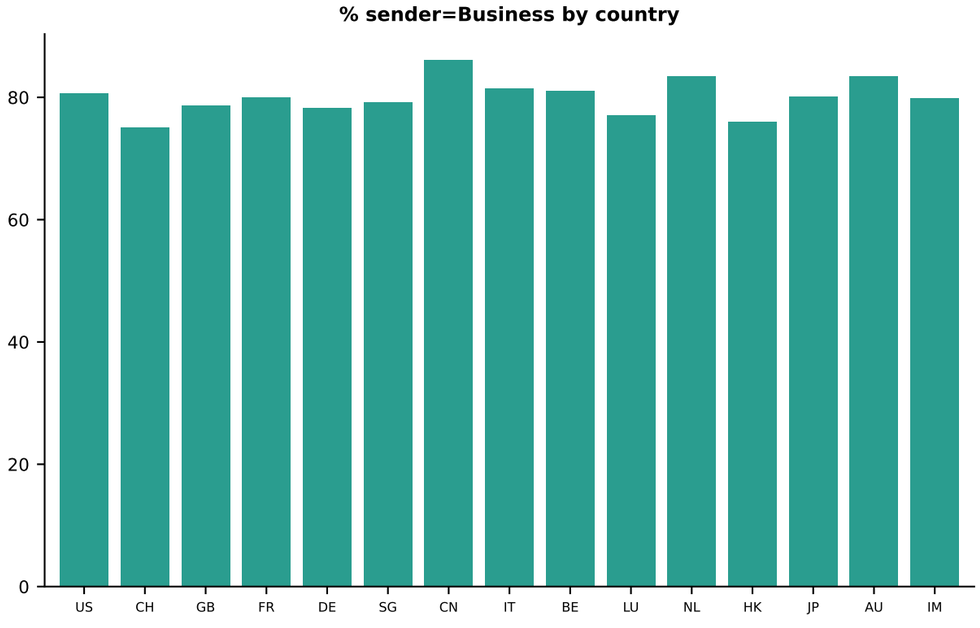
Source: Swift artificial payments dataset (n=10,000,000). All charts computed on the full data.

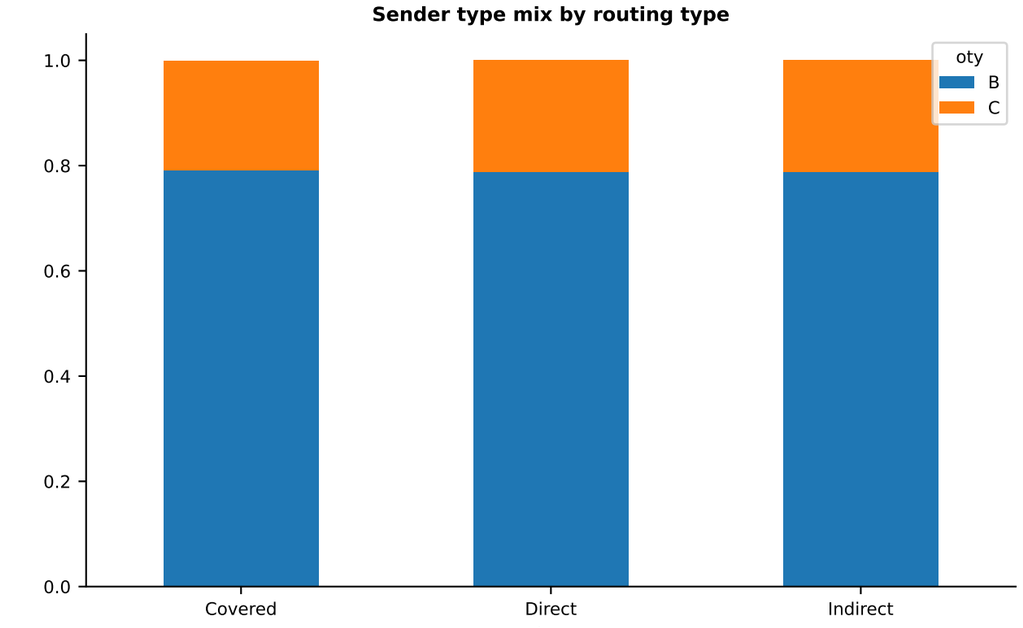
8. Party types

RULE / HYPOTHESIS: H: B/C mix depends on amount & corridor; consumer-to-consumer payments are small; B->B large.





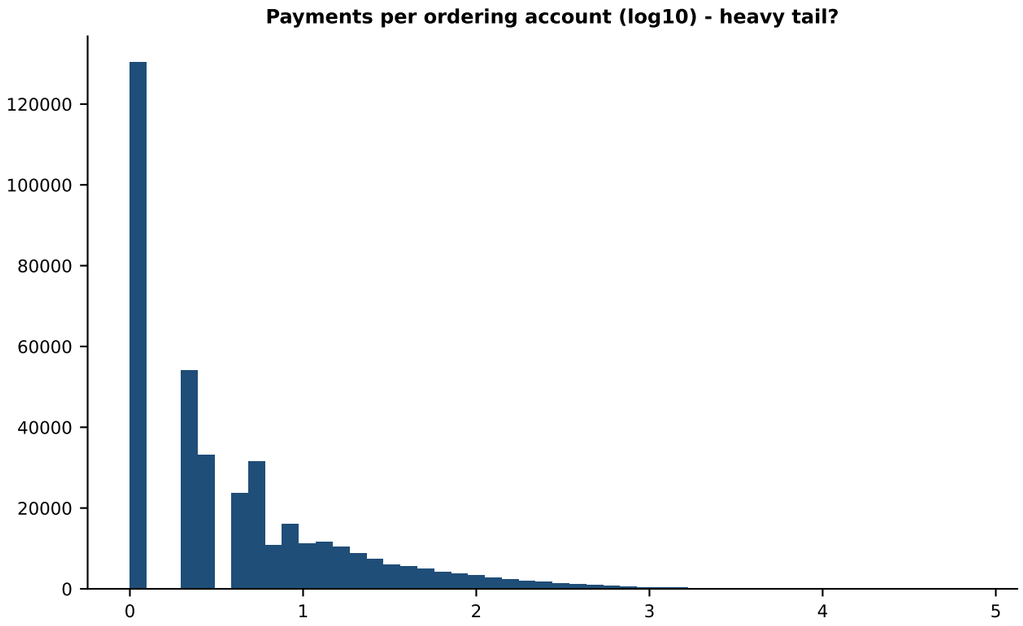


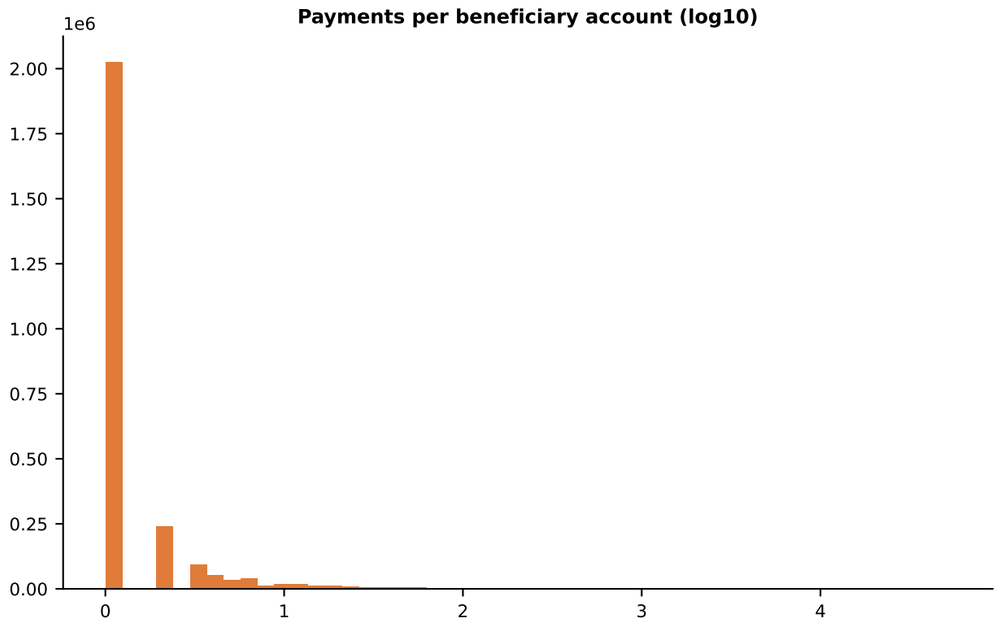


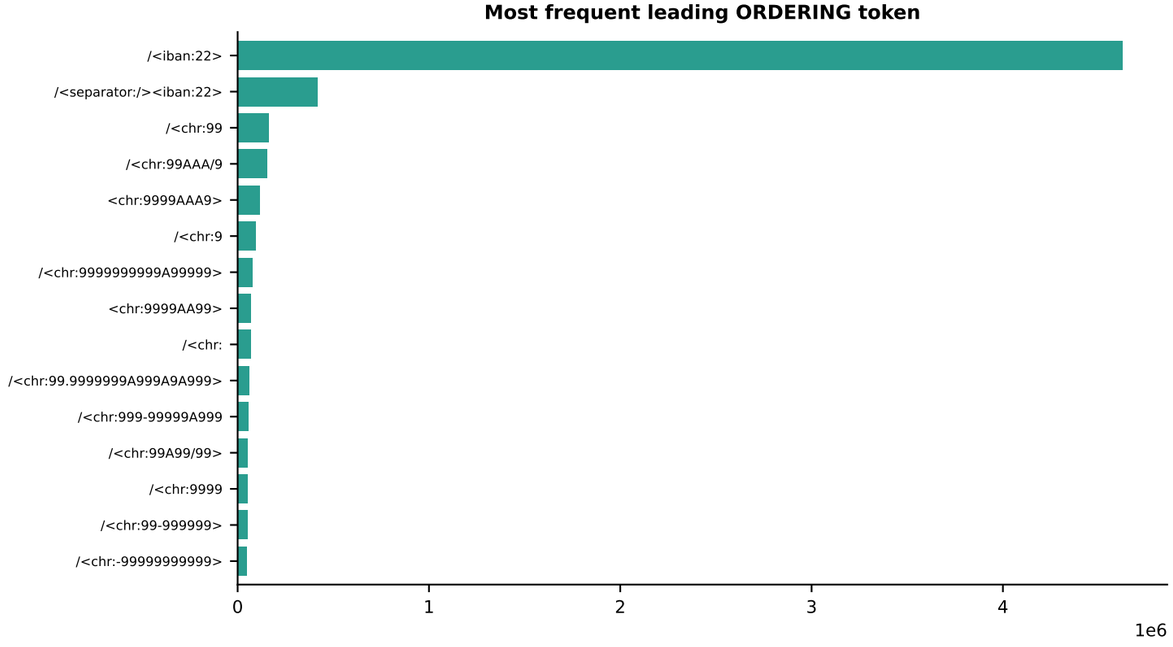
Source: Swift artificial payments dataset (n=10,000,000). All charts computed on the full data.

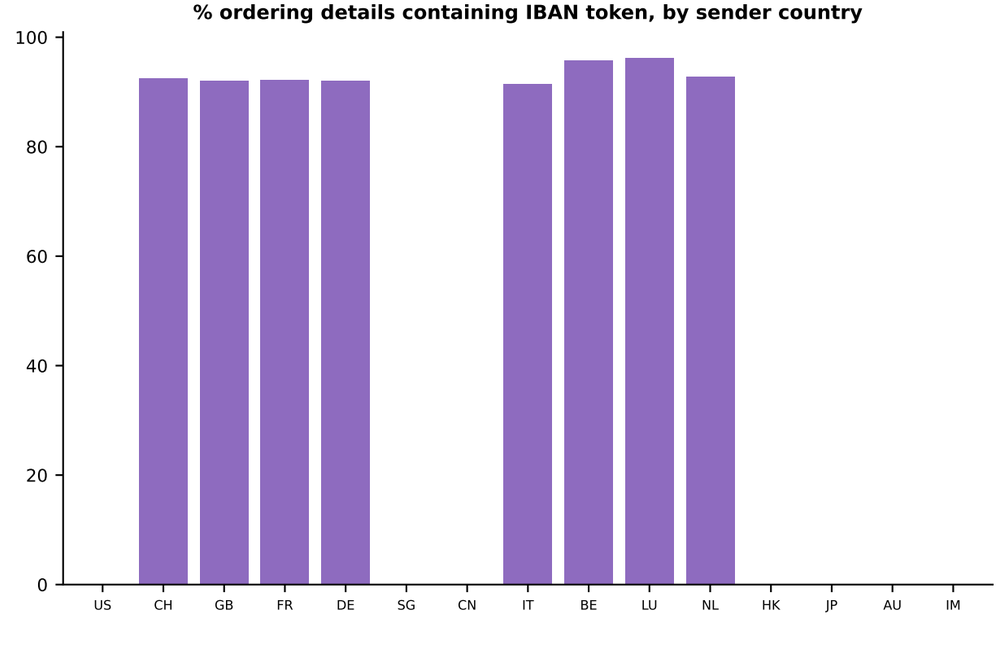
9. Party details & accounts

RULE / HYPOTHESIS: H: Account reuse follows a power law; tokens (IBAN, country code) depend on country; detail text mirrors BIC country.





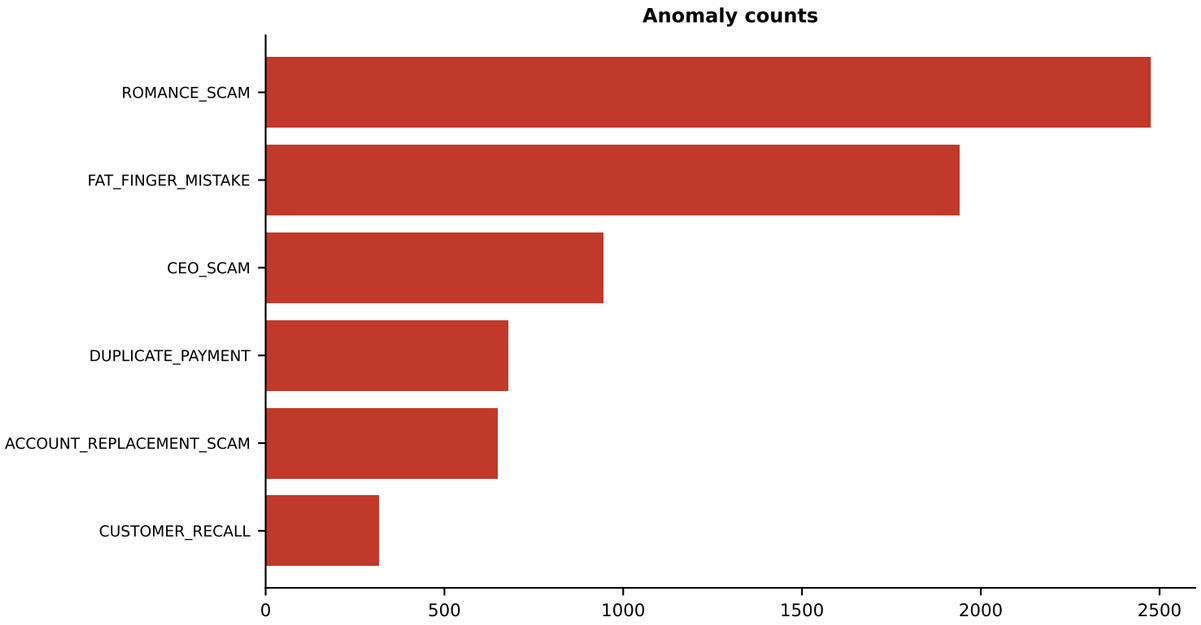


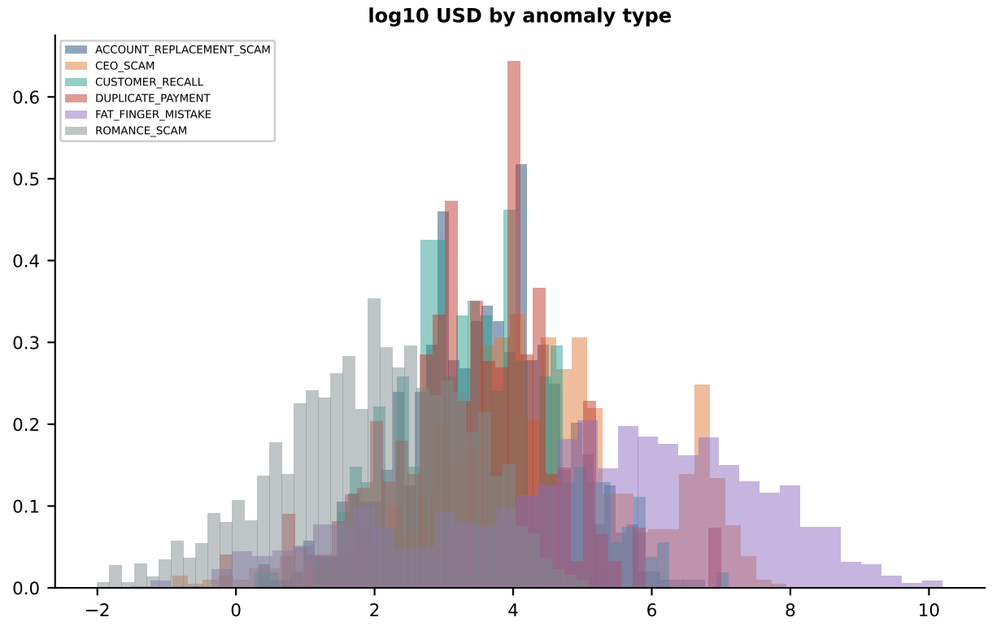


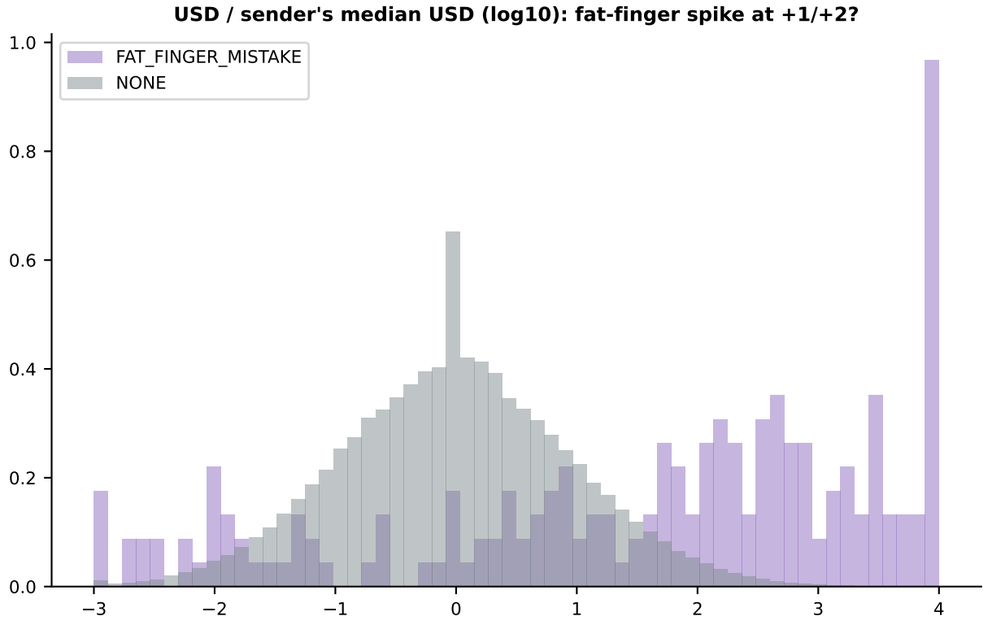
Source: Swift artificial payments dataset (n=10,000,000). All charts computed on the full data.

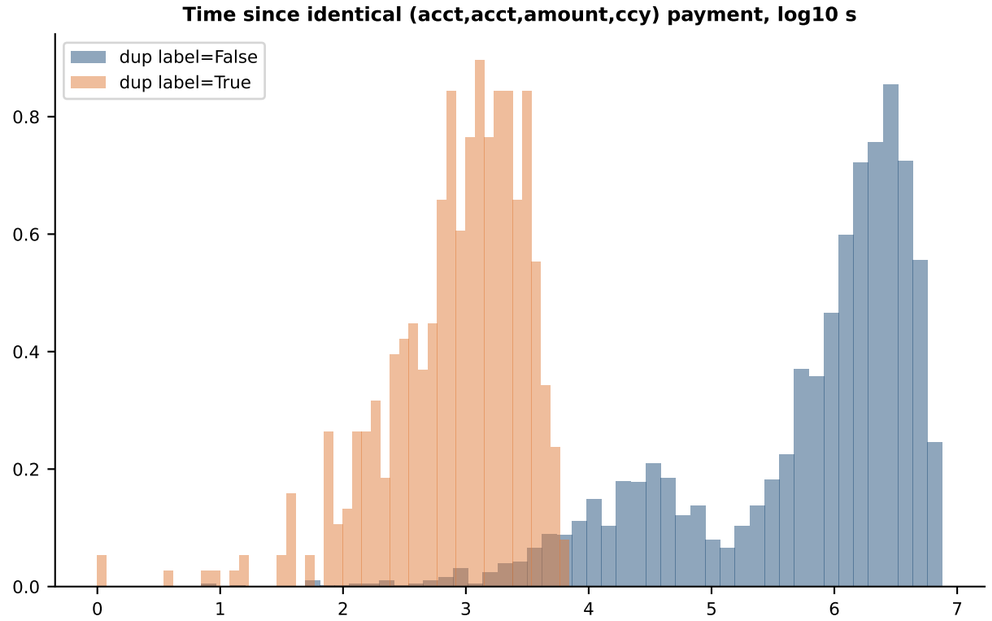
10. Anomalies

RULE / HYPOTHESIS: H: Each ANOMALY_TYPE has a signature: FAT_FINGER = amount x10/x100 vs sender norm; DUPLICATE = same acct/amount repeated in short window; ROMANCE = repeated C->C to one beneficiary; CEO = B, large, new beneficiary; ACCOUNT_REPLACEMENT = known beneficiary BIC, new account.









Source: Swift artificial payments dataset (n=10,000,000). All charts computed on the full data.